



Science

Stage 9

Paper 1

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

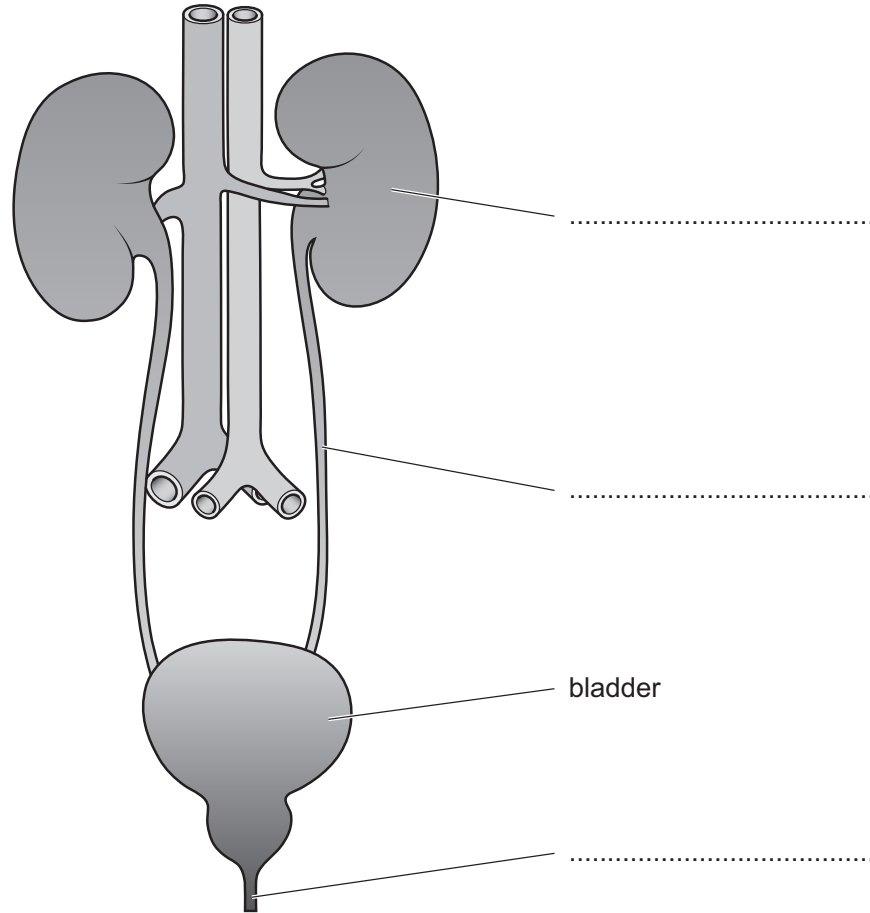
INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Look at the diagram of the human excretory (renal) system.



- (a) Complete the labels on the diagram.

One has been done for you.

[3]

- (b) What is the function of the human excretory (renal) system?

.....
 [1]

2 The table shows information about elements.

atomic symbol	number of electrons in	
	one atom of the element	one ion of the element
Li	3	2
Mg	12	10
Al	13	10
Cl	17	18
K	19	18
Ca	20	18

(a) (i) Which atom loses three electrons to form an ion?

Circle the correct answer.

Li Mg Al Cl K Ca [1]

(ii) Which atom forms a negative ion?

.....

Explain your answer.

.....

.....

[2]

(iii) Two elements in the table are in Group 1.

Write down the atomic symbols of these **two** elements.

Use the Periodic Table on page 18 to help you.

..... and

[1]

- (b) Some elements make compounds with ionic bonds.

Describe what is meant by the words **ionic bond**.

.....

.....

..... [2]

- 3 This question is about heat and temperature.

- (a) Describe the difference between heat and temperature.

.....

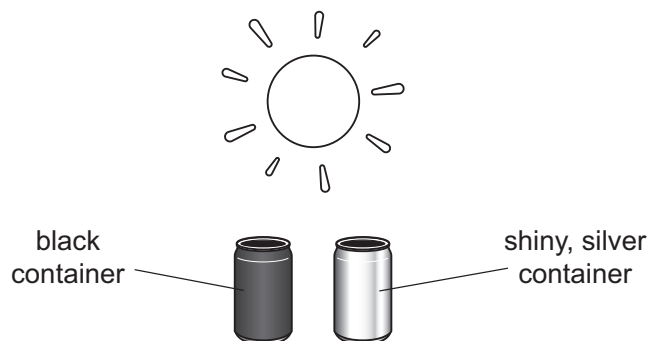
.....

..... [2]

- (b) Rajiv investigates how the temperature of cold water increases in two different metal containers.

He puts the same volume of cold water in each container.

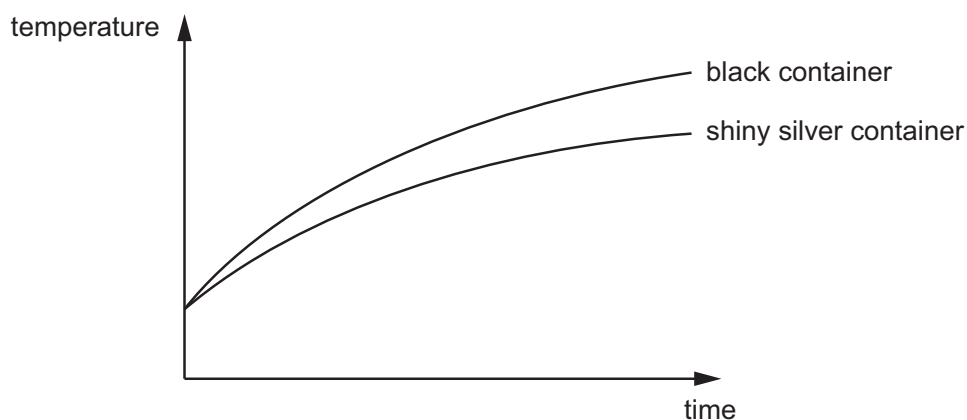
He puts the two containers in the sun.



- (i) Name the equipment Rajiv uses to measure the temperature of the water.

..... [1]

- (ii) Look at the graph showing how the temperature of the water in each container increases with time.



The black container gains the most thermal energy.

Explain how you know from the graph.

.....
 [1]

- (iii) Explain why the black container gains more thermal energy by radiation.

.....
 [1]

- (c) Explain how the thermal energy travels **through** the metal containers.

Complete these sentences.

Thermal energy travels through the metal containers by the process of

The particles in the metal gain thermal energy and more.

As energy passes through the metal the particles with each other.

[2]

- 4 There are different theories for the formation of the Moon.

Tick (✓) **one** correct statement about the **collision theory** for the formation of the Moon.

The Moon formed in another part of the Solar System and was later captured by Earth's gravity.

☐

The Moon was formed at the same time as the Earth was formed.

☐

The Moon formed when Earth hit another smaller planet, the debris collected in an orbit around the Earth.

☐

The Earth was spinning so fast that some material broke off and began to orbit the Earth.

☐

[1]

5 Mia investigates displacement reactions.

In her first experiment she:

- puts some iron sulfate solution into a test-tube
- adds a piece of zinc to the iron sulfate solution
- records if a displacement reaction takes place.

Mia repeats the experiment, but this time uses different metals.

(a) Look at table showing Mia's results from the first experiment.

metal	is there a displacement reaction?
zinc	yes
magnesium	
copper	
iron	

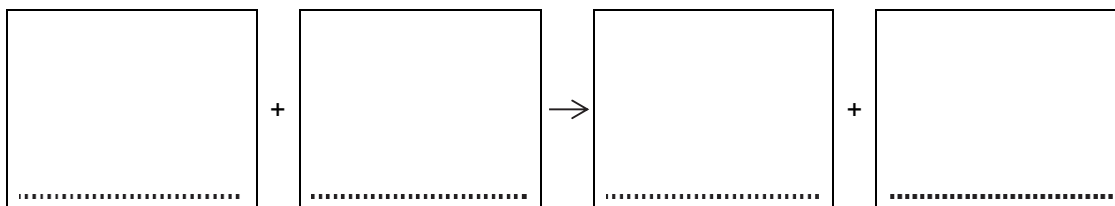
Predict if a displacement reaction takes place with the other metals.

Write your predictions in the table.

[1]

(b) Complete the word equation for the reaction between zinc and iron sulfate solution.

Look at the results.



[2]

(c) Mia replaces the iron sulfate solution each time with a new sample.

Explain why Mia does **not** use the same sample of iron sulfate solution each time.

.....
 [1]

- 6 The diagram shows the pathway of water from the roots to a leaf in a flowering plant.



- (a) (i) Name the cells in the root that absorb water from the soil.

..... [1]

- (ii) Name the tube in the stem that transports water up the flowering plant.

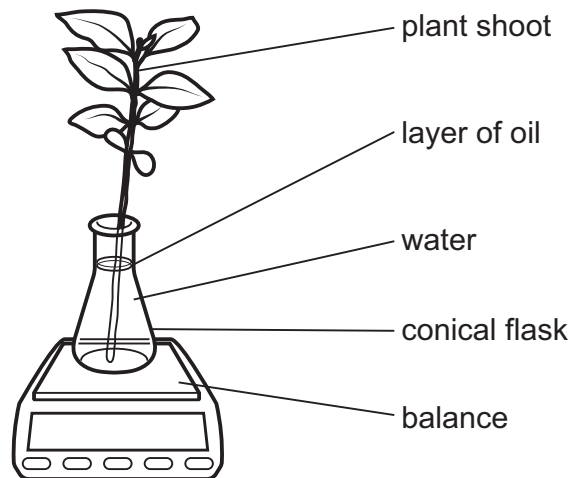
..... [1]

- (iii) Name the process that causes the loss of water from the surface of the leaf.

..... [1]

- (b) Carlos investigates water loss from a plant shoot.

Look at the equipment he uses.



At the start, Carlos measures the mass of the flask, its contents and the plant shoot.

After 24 hours he measures the mass of the flask, its contents and the plant shoot.

The change in mass is the mass of the water lost from the plant shoot.

Here are his results.

mass at start in g	mass after 24 hours in g	mass of water lost in g
820	811	

- (i) Calculate the mass of water lost by the plant shoot.

Write your answer in the table.

[1]

- (ii) Oil is waterproof.

Suggest why Carlos places a layer of oil on top of the water in the conical flask.

.....

..... [1]

- (iii) Carlos wants to improve his experiment.

He repeats his experiment using a different balance.

Here are his results.

mass at start in g	mass after 24 hours in g	mass of water lost in g
820.3	811.4	8.9

Carlos thinks these results are better.

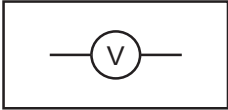
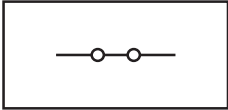
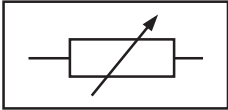
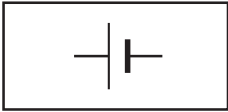
Explain why Carlos is correct.

.....

..... [1]

- 7 Draw a straight line from the **name** of the component to its correct **symbol** and **function**.

One has been done for you.

symbol	name	function
	voltmeter	the source of energy in the circuit
	switch	to measure the voltage across a component
	cell	to change the current in a circuit
	variable resistor	to turn a circuit on or off

[3]

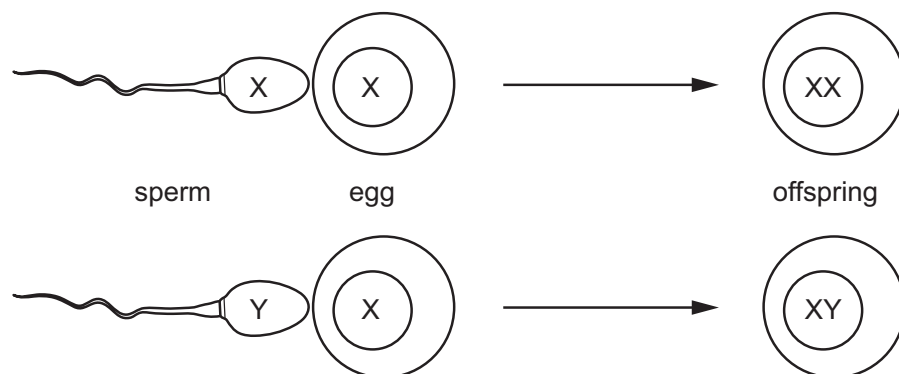
- 8 Offspring inherit characteristics from their parents when a sperm and an egg fuse together.

(a) Name the process that happens when a sperm and an egg fuse together.

..... [1]

(b) The sperm and the egg that fuse together determine the sex of the offspring.

Look at the diagram showing the chromosomes in the sperm and the egg which determine the sex of the offspring.



Explain why there is a 50% chance of the offspring being male.

Use the diagram to help you.

.....

 [2]

(c) The offspring produced look similar but **not** identical to their parents.

Explain why the offspring produced will **not** look identical to their parents.

.....

 [1]

9 Sixty-five million years ago an asteroid collided with Earth creating a large crater.

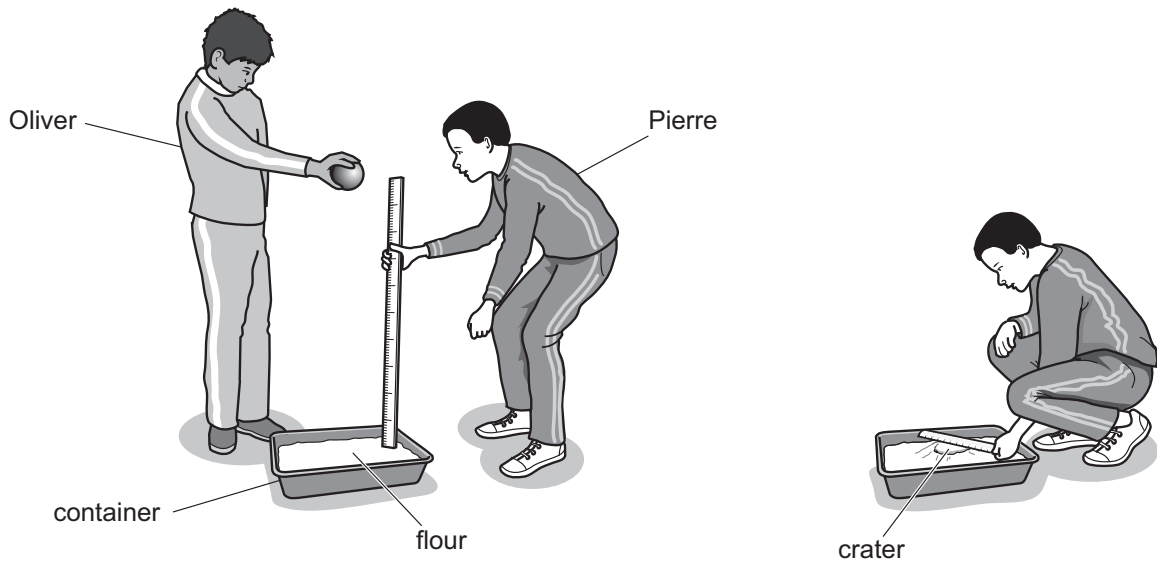
(a) Describe **two other** effects of this asteroid colliding with the Earth.

1

2

[2]

(b) Oliver and Pierre investigate the effect of the size of asteroids on the diameter of craters formed.



Oliver and Pierre:

- measure the diameter of a ball
- drop the ball into 5 cm depth of flour from a height of 1 m
- measure the diameter of the crater formed
- repeat the experiment four times.

Oliver and Pierre test four different sizes of ball.

Here are their results.

diameter of ball in mm	diameter of crater in mm					mean diameter of crater in mm
8	12	11	12	12	12	11.8
10	14	13	12	13	14	13.2
40	37	36	35	36	37	36.2
50	46	48	45	47	48	46.8

Oliver and Pierre are discussing their results.

They think their results are reliable.

Write down **two** reasons why they are correct.

- 1
-
- 2
-

[2]

10 Magnesium reacts with dilute sulfuric acid to make magnesium sulfate solution and hydrogen gas.

(a) Complete these sentences about the rate of this reaction.

The rate of this reaction is increased by increasing the concentration **or**

the of the dilute sulfuric acid.

One other way of increasing the rate of this reaction is to increase

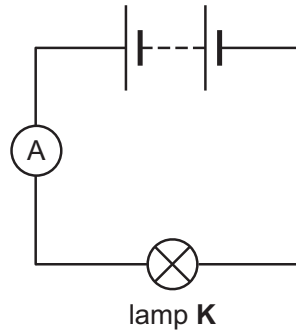
the of the magnesium.

[2]

(b) Describe how solid magnesium sulfate is made from magnesium sulfate solution.

..... [1]

11 Angelique makes an electrical circuit.



(a) The reading on the ammeter is 0.025 A.

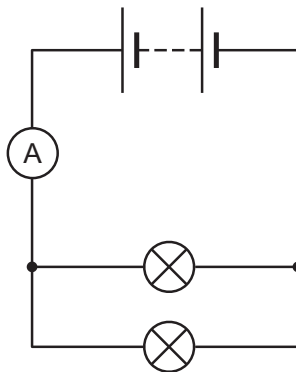
The voltage across lamp **K** is 3.0 V.

Calculate the resistance of lamp **K**.

resistance = Ω [2]

(b) Angelique makes another circuit with two lamps.

Look at the diagram of this circuit.



Describe what happens to the current as it flows through this circuit.

.....

 [2]

12 This is a question about nebulae.

(a) Complete the sentences.

Nebulae are clouds of and

New stars are formed in some nebulae called

[2]

(b) Stars can be classified according to their properties.

Look at the table of the properties of some star types.

star type	colour	relative mean mass (the Sun = 1)	relative mean radius (the Sun = 1)	relative mean luminosity (the Sun = 1)
O	blue	60	15	1 400 000
B	blue	18	7	20 000
A	blue	3.2	2.5	80
F	blue to white	1.7	1.3	6
G	white to yellow	1.1	1.1	1.2
K	orange to red	0.8	0.9	0.4
M	red	0.3	0.4	0.04

Look at the key used to identify a star type.

Identify the star types of the five stars **Antares**, **Arcturus**, **Capella**, **Lactera** and **Sirius**.

Write your answers in the correct star type in the key.

Lactera has been done for you.

			star type
1	star colour is blue or blue to white	go to 2	
	star colour is not blue or blue to white	go to 3	
2	mean mass is 60 times greater than the mass of the Sun	Lactera	O
	mean mass is 3.2 times greater than the mass of the Sun	Sirius	
3	mean mass is greater than the mass of the Sun	Capella	
	mean mass is less than the mass of the Sun	go to 4	
4	mean radius is about half that of the Sun	Antares	
	mean luminosity is about half that of the Sun	Arcturus	

[2]

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The Periodic Table of Elements

Group																							
1	2											3	4	5	6	7	8						
												1 H hydrogen 1											2 He helium 4
		</																					



Science

Stage 9

Paper 1

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

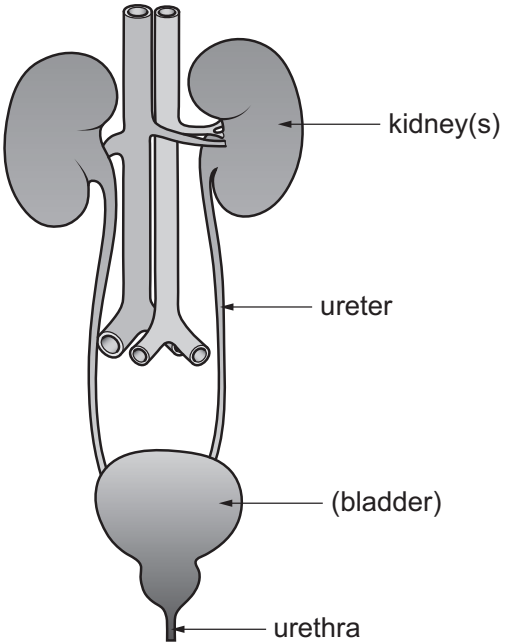
Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	 The diagram illustrates the human urinary system. It shows two kidneys at the top, connected by a central vertical tube. From each kidney, a ureter descends to a bladder-like structure at the bottom. A single urethra exits the bottom of the bladder. Labels with arrows point to the kidney(s), ureter, (bladder), and urethra.	3	each correct answer = 1 mark
1(b)	filters the blood or removes urea from the blood	1	Ignore cleans the blood Accept removes waste products from the blood Accept correct named waste products

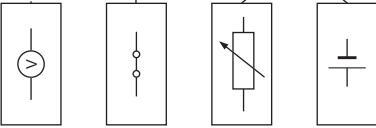
Question	Answer	Marks	Further Information
2(a)(i)	Al	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
2(a)(ii)	Cl gains an electron	2	each correct answer = 1 mark Accept (ion) has 1 more electron than the atom
2(a)(iii)	Li and K	1	both needed in any order for the mark
2(b)	attraction between ions (attraction between) oppositely charged ions or positive and negative ions	2	each correct answer = 1 mark Accept electrostatic attraction between ions = 2 marks

Question	Answer	Marks	Further Information
3(a)	heat is the amount of (thermal) energy in an object temperature is how hot an object is or temperature is how cold an object is	2	each correct answer = 1 mark
3(b)(i)	thermometer	1	Accept temperature probe
3(b)(ii)	(idea that) the temperature has increased the most	1	Accept the black container has reached the highest temperature Accept shiny silver container increased the least Accept (except at start) the temperature is always higher (at the same time)
3(b)(iii)	black (container) is a better absorber (of radiation)	1	Accept shiny or silver object is a worse absorber Accept black (container) reflects less radiation Note answer must be comparative
3(c)	Thermal energy travels through the metal container by conduction . The particles in the metal gain thermal energy and vibrate more. As energy passes through the metal the particles collide with each other.	2	all three correct = 2 marks one or two correct = 1 mark Accept to and fro or move back and forth Accept collide for vibrate Accept hit or bump or knock

Question	Answer	Marks	Further Information
4	The Moon formed in another part of the Solar System and was later captured by Earth's gravity. ☐ The Moon was formed at the same time as the Earth was formed. ☐ The Moon formed when Earth hit another smaller planet, the debris collected in an orbit around the Earth. ☒ The Earth was spinning so fast that some material broke off and began to orbit the Earth. ☐	1	more than one box ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

Question	Answer	Marks	Further Information										
5(a)	<table><tr><td>metal</td><td>is there a displacement reaction?</td></tr><tr><td>zinc</td><td>(yes)</td></tr><tr><td>magnesium</td><td>yes</td></tr><tr><td>copper</td><td>no</td></tr><tr><td>iron</td><td>no</td></tr></table>	metal	is there a displacement reaction?	zinc	(yes)	magnesium	yes	copper	no	iron	no	1	all three correct = 1 mark
metal	is there a displacement reaction?												
zinc	(yes)												
magnesium	yes												
copper	no												
iron	no												
5(b)	<table><tr><td>zinc</td><td>+</td><td>iron sulfate</td><td>→</td><td>zinc sulfate</td><td>+</td><td>iron</td></tr></table>	zinc	+	iron sulfate	→	zinc sulfate	+	iron	2	correct reactants in any order = 1 mark correct products in any order = 1 mark			
zinc	+	iron sulfate	→	zinc sulfate	+	iron							
5(c)	(idea that) iron sulfate is used up in the first reaction	1											

Question	Answer	Marks	Further Information
6(a)(i)	root <u>hair</u> (cells)	1	
6(a)(ii)	xylem	1	
6(a)(iii)	transpiration	1	Accept diffusion or evaporation
6(b)(i)	9 (g)	1	Accept answer written near question, but answer in table takes precedence
6(b)(ii)	(idea of) to stop water loss (from the surface of the water in the conical flask) or to stop evaporation (of water from the conical flask)	1	Do not accept to stop water loss from the plant shoot or evaporation from the plant shoot
6(b)(iii)	the balance is more precise	1	Accept the results are more precise Do not accept results are more accurate

Question	Answer	Marks	Further Information
7	symbol  name voltmeter switch cell variable resistor function the source of energy in the circuit to measure the voltage across a component to change the current in a circuit to turn a circuit on or off <i>(Note: In the original image, dashed lines connect voltmeter to 'to measure the voltage across a component', switch to 'to turn a circuit on or off', cell to 'the source of energy in the circuit', and variable resistor to 'to change the current in a circuit'.)</i>	3	all six lines correct = 3 marks four or five lines correct = 2 marks two or three lines correct = 1 mark one correct = 0 marks if two lines to (or from) one box and one is incorrect, then both lines are incorrect

Question	Answer	Marks	Further Information
8(a)	fertilisation	1	
8(b)	any two from half the sperm has Y (chromosome) or half the sperm has X (chromosome) if the sperm which has a Y chromosome fertilizes the egg the offspring will be male if the sperm which has an X chromosome fertilizes the egg the offspring will be female	2	each correct answer = 1 mark Accept so 50% offspring are male because XY Accept so 50% offspring are female because XX
8(c)	(idea that) offspring inherit genes from both parents	1	Ignore offspring inherit different chromosomes

Question	Answer	Marks	Further Information
9(a)	any two from mass extinction (of dinosaurs/animals) climate change or long term cooling of the Earth's temperature tsunami wildfires	2	each correct answer = 1 mark Accept (idea of) increase in temperature near impact
9(b)	any two from all their results for each ball are similar the mean for each ball is similar to their results there are no anomalous results	2	each correct explanation = 1 mark

Question	Answer	Marks	Further Information
10(a)	The rate of this reaction is increased by increasing the concentration or the temperature of the dilute sulfuric acid One other way of increasing the rate of this reaction is to increase the surface area of the magnesium	2	each correct sentence = 1 mark Ignore concentration
10(b)	evaporation	1	Accept crystallisation or heating Do not accept filtration or distillation or sieving

Question	Answer	Marks	Further Information
11(a)	(resistance) = $\frac{\text{voltage}}{\text{current}} = \frac{V}{I} = \frac{3.0}{0.025}$ 120 (Ω)	2	correct formula or evidence of correct substitution = 1 mark Accept 120 with or without working = 2 marks Ignore incorrect unit
11(b)	any two from this is a parallel circuit (idea that) current divides at the junction (idea that) current joins up after going through the lamps	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information																																
12(a)	Nebulae are clouds of dust and gas New stars are formed in some nebulae called stellar nurseries	2	each correct sentence = 1 mark Accept dust and gas in either order Accept hydrogen or helium for gas																																
12(b)	<table border="1"> <thead> <tr> <th></th><th></th><th></th><th>star type</th></tr> </thead> <tbody> <tr> <td rowspan="2">1</td><td>star colour is blue or blue to white</td><td>go to 2</td><td></td></tr> <tr> <td>star colour is not blue or blue to white</td><td>go to 3</td><td></td></tr> <tr> <td rowspan="2">2</td><td>mean mass is 60 times greater than the mass of the Sun</td><td>Lactera</td><td>(O)</td></tr> <tr> <td>mean mass is 3.2 times greater than the mass of the Sun</td><td>Sirius</td><td>A</td></tr> <tr> <td rowspan="2">3</td><td>mean mass is greater than the mass of the Sun</td><td>Capella</td><td>G</td></tr> <tr> <td>mean mass is less than the mass of the Sun</td><td>go to 4</td><td></td></tr> <tr> <td rowspan="2">4</td><td>mean radius is about half that of the Sun</td><td>Antares</td><td>M</td></tr> <tr> <td>mean luminosity is about half that of the Sun</td><td>Arcturus</td><td>K</td></tr> </tbody> </table>				star type	1	star colour is blue or blue to white	go to 2		star colour is not blue or blue to white	go to 3		2	mean mass is 60 times greater than the mass of the Sun	Lactera	(O)	mean mass is 3.2 times greater than the mass of the Sun	Sirius	A	3	mean mass is greater than the mass of the Sun	Capella	G	mean mass is less than the mass of the Sun	go to 4		4	mean radius is about half that of the Sun	Antares	M	mean luminosity is about half that of the Sun	Arcturus	K	2	all four correct = 2 marks two or three correct = 1 mark one correct = 0 marks
			star type																																
1	star colour is blue or blue to white	go to 2																																	
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Science

Stage 9

Paper 2

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

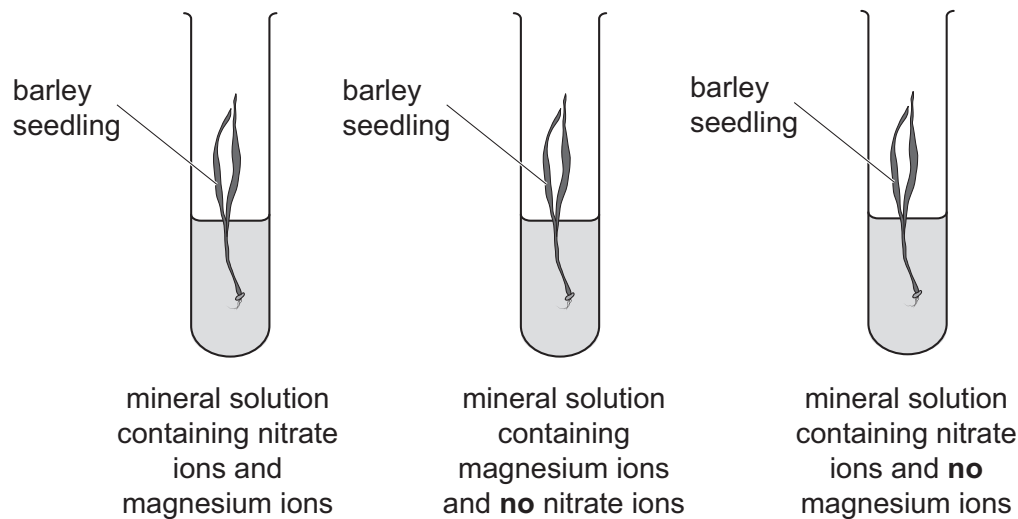
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Priya investigates the effect of nitrate ions and magnesium ions on the growth of barley seedlings.

Priya does three experiments.

In each experiment she puts one barley seedling and 3 cm^3 of a mineral solution into a test-tube.



After four weeks Priya dries the seedlings in an oven and measures their dry mass.

Here are her results.

experiment	dry mass of the barley seedling in g
mineral solution containing nitrate ions and magnesium ions	3.8
mineral solution containing magnesium ions and no nitrate ions	1.9
mineral solution containing nitrate ions and no magnesium ions	2.8

Describe and explain the effect of nitrate ions and the effect of magnesium ions on the growth of barley seedlings.

Use information from the table in your answer.

.....

.....

.....

.....

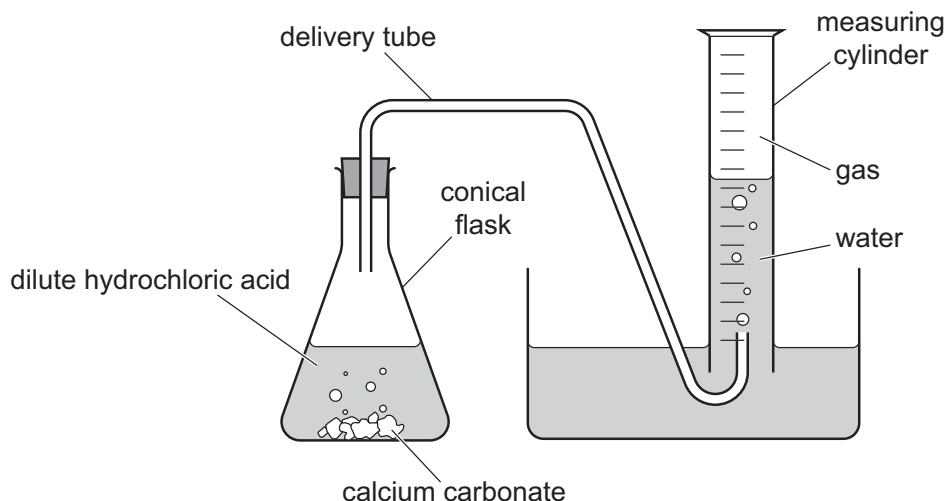
.....

[4]

2 Dilute hydrochloric acid reacts with calcium carbonate.

Blessy plans an investigation to find out the effect of concentration of acid on the rate of this reaction.

The diagram shows the equipment she uses.



(a) (i) Identify the independent variable in this investigation.

..... [1]

(ii) Describe how Blessy measures the dependent variable in this investigation.

dependent variable

how it is measured

.....

[1]

(iii) Identify **two** variables that Blessy must control.

1

2

[1]

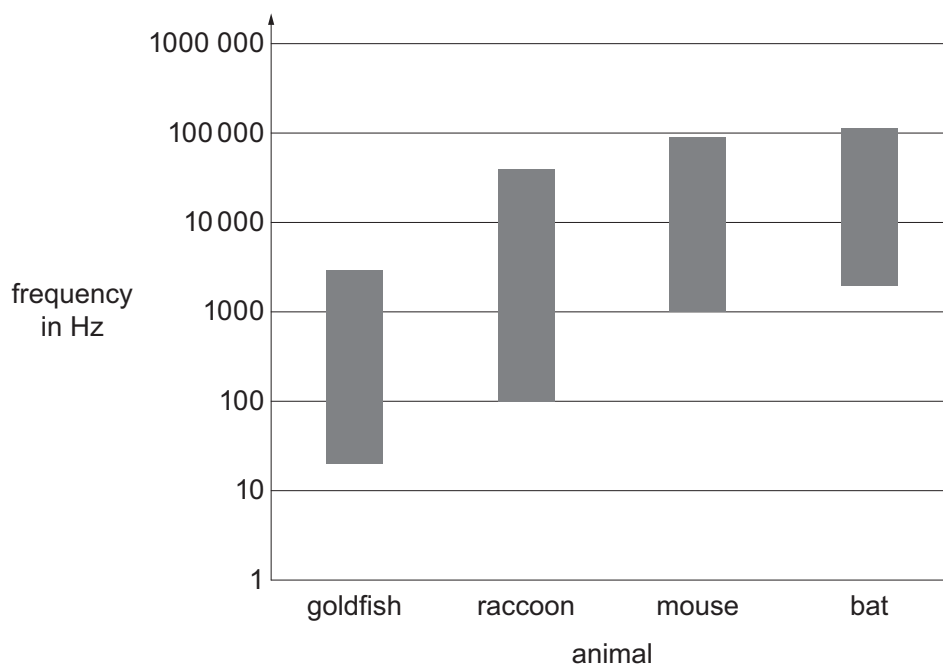
(b) Identify **one** safety risk in the investigation and describe how to control the risk.

safety risk

how to control the risk

[2]

- 3 The diagram shows the range of frequencies of sound waves that different animals hear.



- (a) Write down the **lowest** frequency of sound a raccoon hears.

..... Hz

[1]

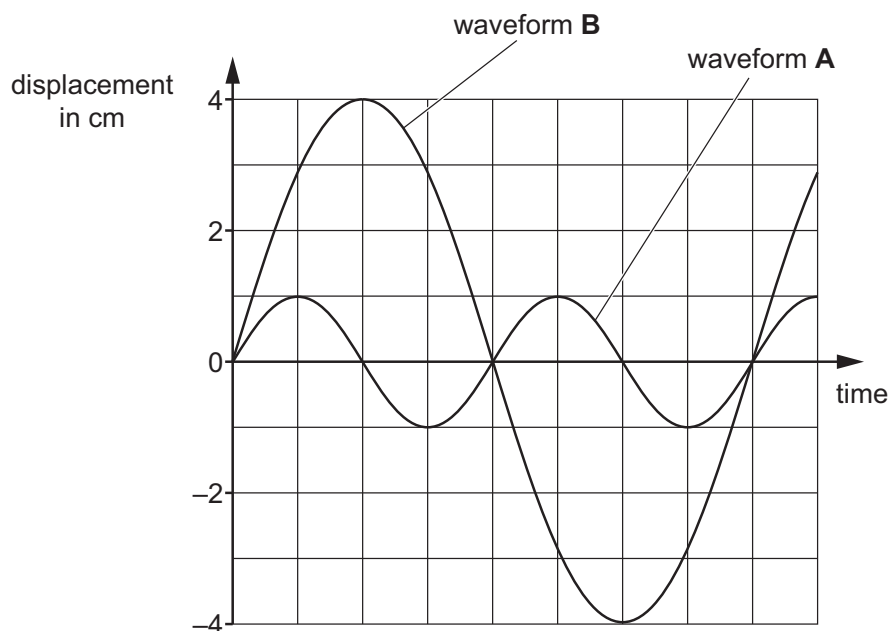
- (b) Estimate the range of frequencies a mouse hears.

..... Hz

[1]

(c) Look at the graph showing the waveforms of two sound waves.

Waveform **A** and waveform **B** have different frequencies and different amplitudes.



(i) Describe how increasing the frequency of a sound wave changes the sound we hear.

..... [1]

(ii) Describe how increasing the amplitude of a sound wave changes the sound we hear.

..... [1]

(d) (i) Calculate how many times bigger the **amplitude** of waveform **B** is than waveform **A**.

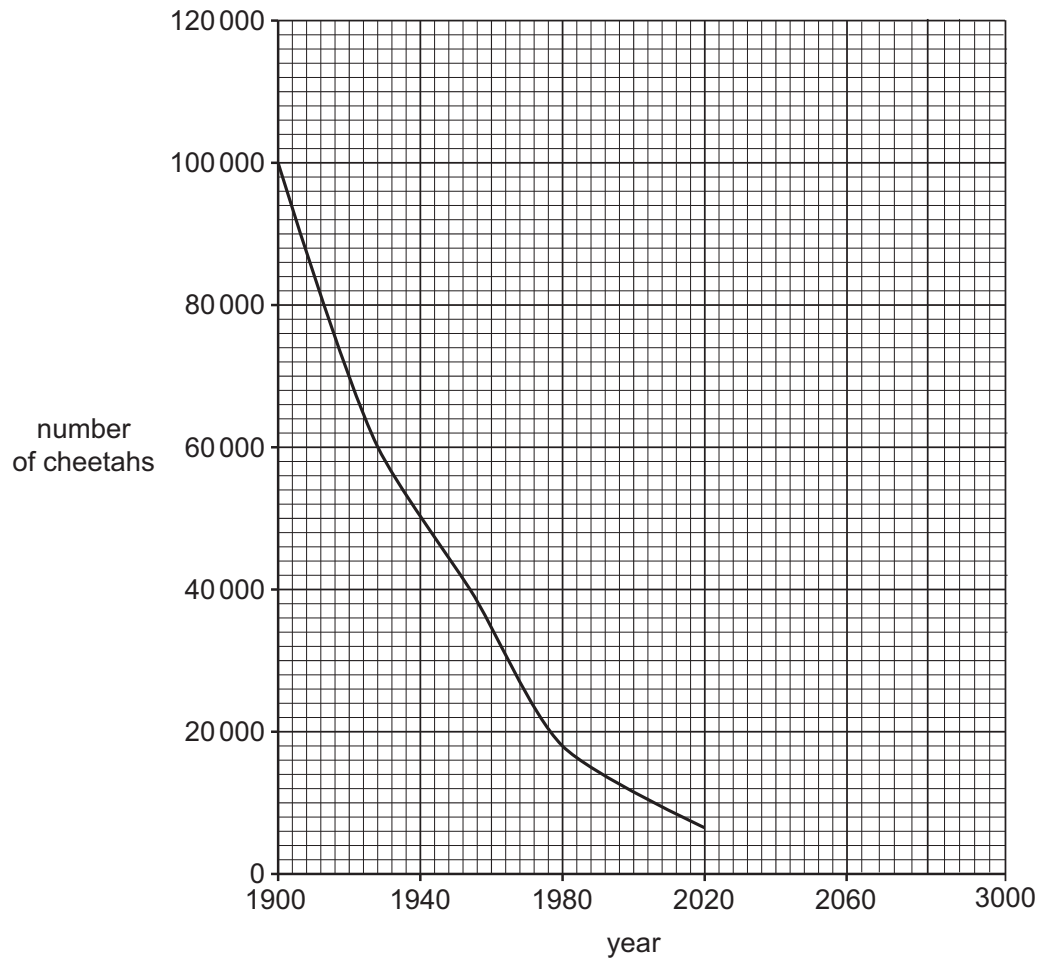
..... [1]

(ii) Calculate how many times bigger the **frequency** of waveform **A** is than waveform **B**.

..... [1]

4 This question is about the population of cheetahs and sea turtles.

(a) Look at the graph showing the change in the number of cheetahs since 1900.



(i) Describe the change in the number of cheetahs between 1900 and 2020.

.....
 [1]

(ii) Draw an extension to the line of best fit to the x-axis.

Estimate the year when the cheetah becomes extinct.

year [2]

- (iii) The change in the number of cheetahs may be due to natural selection.

Describe the theory of natural selection.

.....

.....

.....

..... [2]

- (b) Carbon dioxide levels in the atmosphere are linked to increased Earth surface temperature and rising sea levels.

Sea turtles live in the ocean and come to shore to build nesting sites in the sand.

The sex of the sea turtle offspring is linked to the surface temperature of the sand.

At warmer temperatures more of the offspring are female than male.

- (i) The population of the sea turtles might change because of an increase in the **surface temperature** of **sand**.

Complete the sentences.

The population of sea turtles might **increase** because

.....

The population of sea turtles might **decrease** because

.....

[2]

- (ii) Suggest how the **rising sea level** will affect the population of the sea turtles.

Write down **two** reasons for your answer.

effect on the population

reason 1

.....

reason 2

.....

[2]

- 5 Chen measures the mass and volume of some substances.

He calculates the density of each substance.

The table shows his results.

substance	mass in g	volume in cm ³	density in g/cm ³
A	395	50	7.9
B	0.22	100	0.0022
C	452	40	11
D	328	45	7.3
E	340	38	

- (a) Calculate the density of substance E.

Give your answer to **two** significant figures.

density of substance E = g/cm³ [3]

- (b) Which substance in the table is a gas?

Explain your answer.

substance

explanation

.....

[2]

- 6 Perspiration is a watery liquid produced by the skin.

The water evaporates to cool the skin.

Explain the cooling effect of evaporation.

Use ideas about particles.

.....

.....

.....

..... [3]

- 7 This question is about the carbon cycle.

(a) Draw a straight line to match the **process** to its correct **description**.

process	description
combustion	carbohydrate moves from one organism to another organism
decomposition	the breakdown of dead and decaying waste material
feeding	the burning of fossil fuels
photosynthesis	the release of energy from the breakdown of glucose
respiration	the formation of glucose and oxygen by green plants

[4]

(b) Name the process in the carbon cycle that decreases the amount of carbon in the atmosphere.

..... [1]

- 8 The Earth's crust is split into large pieces of rock.

These large pieces of rock float on top of the molten magma in the mantle.

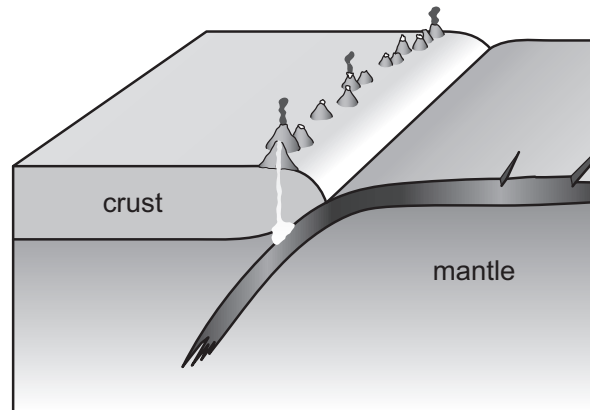
- (a) Write down the name of these large pieces of floating rock.

..... [1]

- (b) Convection currents in the mantle cause these floating rocks to move.

- (i) Look at the diagram showing the formation of volcanoes.

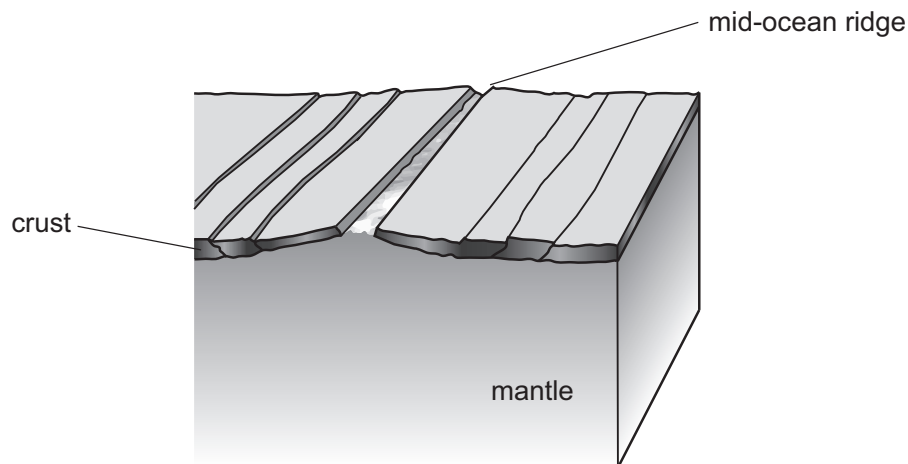
Draw arrows on the diagram to show the pathway of the convection currents in the mantle.



[1]

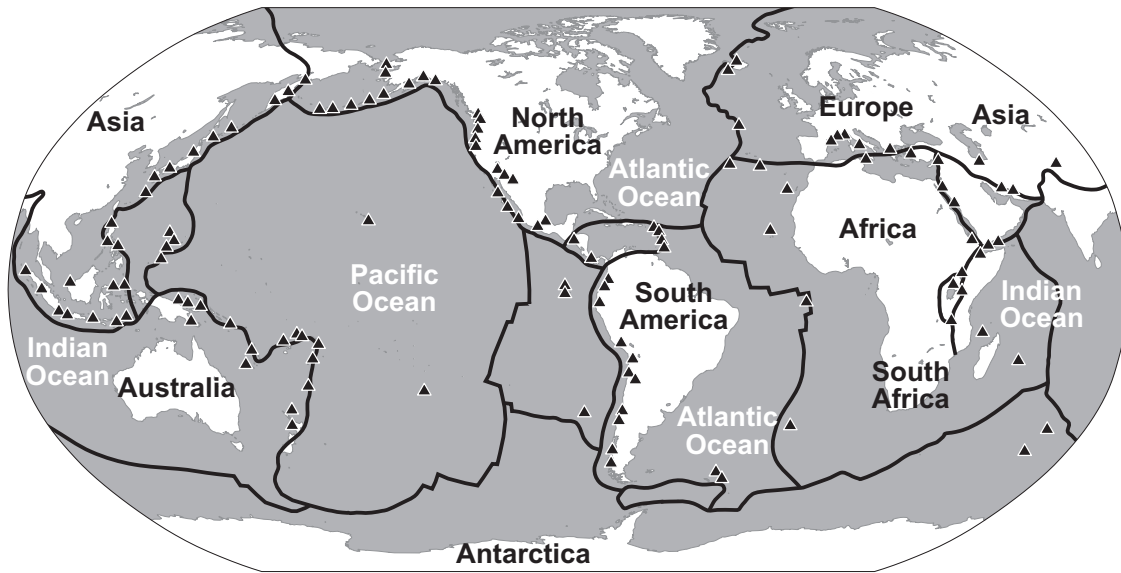
- (ii) Look at the diagram showing the formation of a mid-ocean ridge.

Draw arrows on the diagram to show the pathway of the convection currents which cause the formation of the mid-ocean ridge.



[1]

- (c) Look at the map showing the position of active volcanoes and the boundaries of floating rock in the Earth's crust.



key

▲ active volcano

— boundary of floating rocks

- (i) Why are active volcanoes used as evidence for the location of the boundaries of the floating rocks?

.....
 [1]

- (ii) Suggest why there are **no** active volcanoes near South Africa.

.....
 [1]

9 Copper oxide reacts with dilute nitric acid to make a salt and water.

(a) What is name of the salt made in this reaction?

..... [1]

(b) The mass is conserved during this reaction.

What is the meaning of the words **mass is conserved**?

.....
..... [1]

(c) Water molecules are made in this reaction.

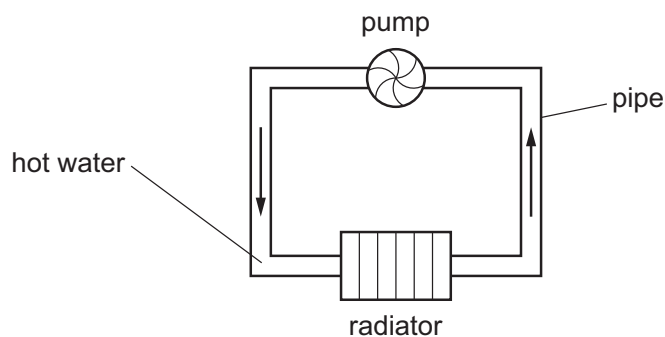
Name and describe the type of bond present in a water molecule.

name

description

.....
[2]

10 Look at the diagram showing a model to help explain how an electrical circuit works.



(a) The radiator in the model represents a lamp in the circuit.

What is represented in the electrical circuit by the pump and the hot water?

pump

hot water

[2]

(b) Write down **one strength** and **one limitation** of this model of an electrical circuit.

strength

.....

limitation

.....

[2]

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Science

Stage 9

Paper 2

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1	when nitrate ions and magnesium ions are both present the (barley seedlings) have the greatest dry mass nitrate ions have a greater effect on the dry mass than magnesium ions / ora nitrate ions are needed to make proteins magnesium ions are needed to make chlorophyll	4	each correct answer = 1 mark Accept the seedling with just nitrate ions grows more than the seedling with just magnesium ions / ora Accept nitrate ions are needed to make DNA or amino acids Accept magnesium ions are needed for photosynthesis

Question	Answer	Marks	Further Information
2(a)(i)	concentration (of acid)	1	
2(a)(ii)	(dependent variable) volume of gas (how it is measured) water level in measuring cylinder	1	both answers correct for the mark
2(a)(iii)	any two from (same) mass (of calcium carbonate) (same) particle size or same surface area (of calcium carbonate) (same) temperature (same) volume (of acid)	1	both answers correct for the mark Do not accept (same) concentration of acid
2(b)	(safety risk) acid is corrosive (how to control the risk) wear eye protection	2	each correct answer = 1 mark Accept acid can burn skin or eyes or acid can irritate skin or eyes or acid can harm skin or eyes Accept wear gloves

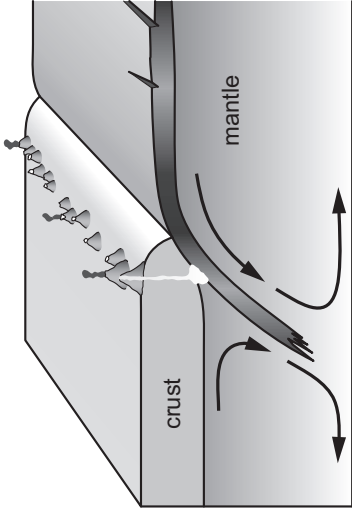
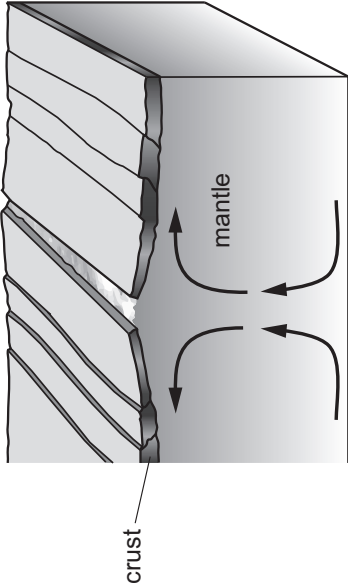
Question	Answer	Marks	Further Information
3(a)	100 (Hz)	1	no tolerance
3(b)	90 000 (Hz) or (range is from) 1000 (Hz) to 90 000 (Hz)	1	Accept between 80 000 and 99 000 (Hz) Accept between 80 000 and 99 000 (Hz) for the value of 90 000 (Hz) no tolerance on the value of 1000 (Hz)
3(c)(i)	increases the pitch	1	Accept higher pitch Note answer must be comparative
3(c)(ii)	increases the loudness or increases the volume	1	Accept louder sound Note answer must be comparative
3(d)(i)	4 (times)	1	
3(d)(ii)	2 (times)	1	Accept waveform A has twice the frequency of waveform B

Question	Answer	Marks	Further Information
4(a)(i)	decreases	1	
4(a)(ii)	line of best fit extended to the x-axis correct year read from the line to the nearest year	2	each correct answer = 1 mark
4(a)(iii)	any two from variation within the species (idea of) best adapted survive or (idea of) competition advantageous adaptations passed on to offspring	2	each correct answer = 1 mark Accept survival of the fittest
4(b)(i)	(increases) there are more females to produce offspring (decreases) there will be fewer males to mate with the females	2	each correct explanation = 1 mark
4(b)(ii)	(effect on the population) decreases (no marks) (reason 1) (idea of) the nesting sites will be flooded (reason 2) the offspring will drown or the offspring will have insufficient oxygen to respire	2	increase = 0 marks for the question each correct reason= 1 mark

Question	Answer	Marks	Further Information
5(a)	correct substitution of numbers into formula or $340 \div 38$ correct division, e.g. 8.947368 correct answer 8.9	3	Accept 8.9 with or without working = 3 marks Accept 8.947368 with or without working = 2 marks correct construction of formula mass $\div$ volume or $340 \div 38$ correct division using their values from substitution Accept truncated answers or incorrect rounding
5(b)	(substance) B (explanation) (idea of) lowest density or density much lower than others or has a very low density	2	each correct answer = 1 mark Accept density of gas usually less than solids or liquids

Question	Answer	Marks	Further Information
6	any three from (water) particles gain energy from skin more energetic (water) particles escape (from surface of skin or to air) less energetic (water) particles remain (on skin) thermal energy transfers from body or skin	3	each correct answer = 1 mark Accept molecules for particles throughout Accept faster molecules escape Accept slower (water) particles remain Accept heat for thermal energy Accept evaporation is endothermic as an extra marking point

Question	Answer	Marks	Further Information
7(a)		4	all five correct = 4 marks three or four correct = 3 marks two correct = 2 marks one correct = 1 mark if two lines from one process to different descriptions = 0 marks for that process
7(b)	photosynthesis	1	

Question	Answer	Marks	Further Information
8(a)	tectonic plates	1	
8(b)(i)		1	arrow heads must be in the correct direction on all arrows drawn
8(b)(ii)		1	arrow heads must be in the correct direction on all arrows drawn Accept a full convection current
8(c)(i)	(idea of) they follow the same pattern or active volcanoes are found where the boundaries of the floating rocks or tectonics plates are located	1	
8(c)(ii)	no boundaries near South Africa	1	

Question	Answer	Marks	Further Information
9(a)	copper nitrate	1	
9(b)	(idea that) mass of reactants is the same as mass of products or no loss or gain of mass during the reaction	1	
9(c)	(name) covalent bond (description) shared pair of electrons	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
10(a)	(pump) cell(s) or battery or power supply (hot water) charge or electrons or current	2	each correct answer = 1 mark Ignore wire
10(b)	any one strength from both water and current are not 'used up' as they pass around the pipes/circuit there is the same amount of water or current at all points same rate of flow of water or current or charge at all points any one limitation from you cannot see what is happening to the water in the pipes it takes time for the water to reach the radiator it takes time for radiator to get hot (but the lamp turns on straight away) water can leak (out of pipes) but current cannot leak (out of wires)	2	each correct answer = 1 mark Accept (idea that) the pump requires energy to work but the cell provides the electrical energy Accept looks like a circuit so easy to visualise Accept to help explain a difficult idea, e.g. current